

Machine Intelligence*Exam*

10.00 - 14.00, 6 January 2023

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Please read the following before solving the exam exercises.

- This exam contains 6 exercises covering different aspects of the course. The exam is worth 100 points in total. Each exercise has been assigned a different amount of points, indicated in brackets at the beginning of the exercise.
- Each exercise has several parts and questions. Please read all of them carefully and check that you have answered all of them.
- The solution has to be written in English.
- This exam is open book. You are allowed to use notes, books, slides, exercise/solution sheets, and the equivalent of a standard pocket calculator. Anything else is considered illegal, in particular, Googling or asking others for help.
- Mobile phones or other devices are not allowed during the exam.
- Before you answer a question make sure that you have read the question carefully. Moreover, when the exercises indicate it, make sure that you argue for your answers (e.g. include intermediate results) so that it is possible to follow your line of thought. Your solutions should be presented in a readable form and written in English.
- You can use the tool that you prefer to answer the exam exercises, but the solutions have to be uploaded to Digital Exam in a single PDF file.
- Please, make sure that your solution is readable, and organized. Remember to indicate which is the question you are answering to.
- In case of emergencies: Students can contact the instructor during the exam by approaching the study secretary

Last but not least, good luck!

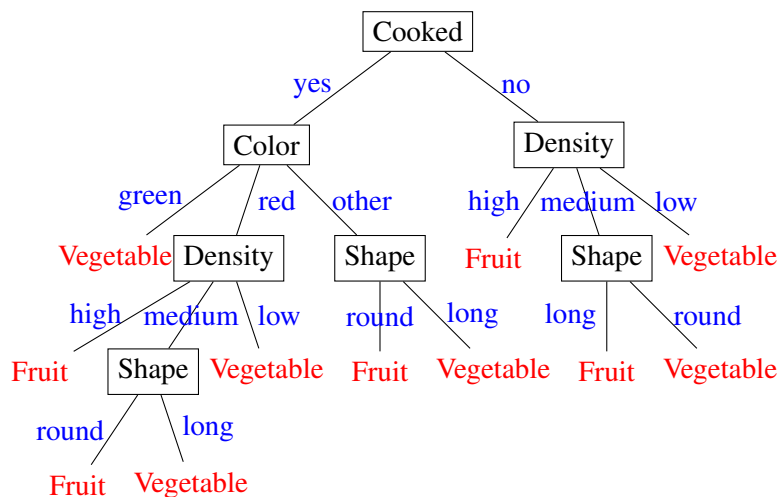
Exercise 1 (22 points)

We want to design a system that is able to classify food to tell us whether is a fruit (positive class) or vegetable (negative class).

	Color	Shape	Density	Cooked	Target class
Banana	Yellow	Long	High	No	Fruit
Lettuce	Green	Round	Low	No	Vegetable
Aubergine	Purple	Long	High	Yes	Vegetable
Carrot	Orange	Long	High	Yes	Vegetable
Green peas	Green	Round	Medium	Yes	Vegetable
Grapes	Green	Round	Medium	No	Fruit
Apple	Red	Round	High	No	Fruit
Zucchini	Green	Long	Medium	Yes	Vegetable

Table 1: Dataset: Test data for parts (i) and (ii) as well as training data for part (iii).

- (i) Consider the following decision tree that has been learned on some data set (not shown here). How does the tree classify each food in the test set from Table 1? Fill the table below to the right.



	Class
Banana	
Lettuce	
Aubergine	
Carrot	
Green peas	
Grapes	
Apple	
Zucchini	

- (ii) Consider now, the following classification that has been computed based on a different tree (not shown here). To measure the quality of the tree, fill the table below:

	Class (Tree ii)	Metric
Banana	Vegetable	True positives
Lettuce	Fruit	False positives
Aubergine	Vegetable	True negatives
Carrot	Vegetable	False negatives
Green peas	Vegetable	Accuracy
Grapes	Fruit	Precision
Apple	Fruit	Recall
Zucchini	Vegetable	

- (iii) Suppose now that we start to learn a new tree, using the dataset in Table 1 as training set. Compute the information gain for the attribute “Shape”. Provide all intermediate results, with a brief explanation of where are they coming from.
- (iv) Compute the information gain for the attribute “Cooked”. Provide all intermediate results, with a brief explanation of where are they coming from.
- (v) Among “Shape” and “Cooked” what attribute would be chosen by the algorithm to split the data? Justify your answer.
- (vi) Suppose that, after learning the new tree (following with the procedure from steps (iii-v) , it has an accuracy, precision, and recall of 1 on the test set of Table 1. Is the new tree then preferable to the one in part (i)? Justify your answer.
- (vii) When we train a Neural Network with Backpropagation, we automatically learn (mark all that apply):
- | | |
|--|---|
| ☐ How many layers to use | ☐ How many neurons to use in each layer |
| ☐ What activation function to use | ☐ The weights in the connections between the neurons |
-

Exercise 2 (13 points)

We have a dataset with characters from a videogame along with their attributes (three numerical attributes), shown below. We want to apply clustering in order to classify all characters into two categories. As distance function, we use the Euclidean distance.

Character ID	#1	#2	#3	#4	#5	#6
Attribute 1 (HP)	25	45	70	40	30	80
Attribute 2 (STR)	2	4	8	3	5	8
Attribute 3 (SPD)	9	4	7	10	9	8

- (i) Suppose that you run k -means using 2 clusters, initializing the algorithm with $C_1 = (20, 1, 1)$, $C_2 = (80, 10, 10)$. At that point, what is the current classification for each of the examples?

Character ID	#1	#2	#3	#4	#5	#6
Class						

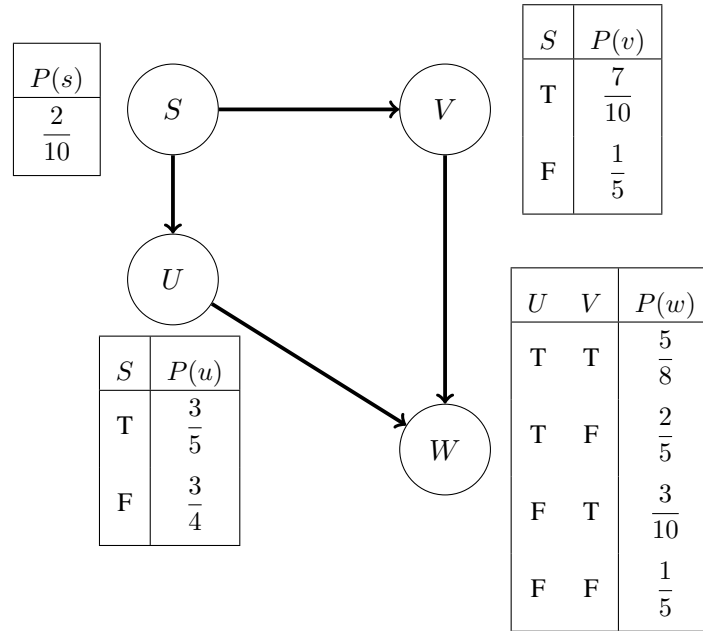
- (ii) What would be the values of C_1 and C_2 after performing one iteration of the k -means algorithm? Explain all steps the algorithm needs to take and the intermediate results you compute.

- (iii) Would the algorithm stop after one iteration? Justify your answer.

- (iv) Do you identify any issue with this data set? If you do, indicate what is the problem and what technique, among the ones considered in the lecture, could you use to improve the results of k -means in this dataset?

Exercise 3 (18 points)

Consider the following Bayesian Network with Boolean random variables S, U, V, W . In what follows, we use the notational conventions from the lecture: For Boolean variable X , x denotes $X = \text{true}$ and $\neg x$ denotes $X = \text{false}$.



- (i) Fill in the following table, answering whether the variables are d-separated or not, for different sets of evidence variables.

	yes	no
S and U are d-separated	☐	☐
S and U are d-separated given W	☐	☐
S and U are d-separated given V	☐	☐
S and U are d-separated given V and W	☐	☐
U and V are d-separated	☐	☐
U and V are d-separated given S	☐	☐
U and V are d-separated given W	☐	☐
U and V are d-separated given S and W	☐	☐
S and W are d-separated	☐	☐
S and W are d-separated given U	☐	☐
S and W are d-separated given V	☐	☐
S and W are d-separated given U and V	☐	☐

Compute the following probabilities. Note: Using tools that automatically compute the results is not allowed, so you must indicate exactly how the result is being computed. In each case, include intermediate steps. In particular, state *which* probabilities provided in the Bayesian Network are combined in order to obtain the result. Then, you may use a calculator to compute the final answer.

(ii) Compute $P(u, \neg v, w, s)$

(iii) Compute $P(\neg u \mid s)$

(iv) Compute $P(s \mid u, w)$

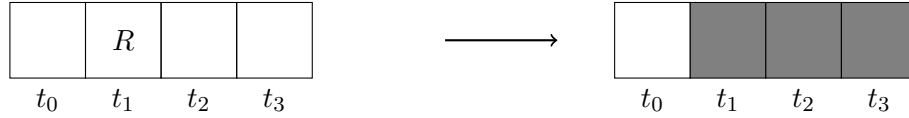
(v) What characterizes a Naïve Bayes model? (mark all that apply):

- | | |
|--|--|
| ☐ The nodes of the Bayesian Network are connected in a specific way | ☐ The probabilities are learned with a small dataset |
| ☐ Assumes the future is independent of the past given the present | ☐ Assumes that the effects of a cause are independent |

Exercise 4 (18 points)

Consider the following planning task. We have a corridor with 4 tiles $t_0, \dots, t_3$. The tiles are placed in a line, i.e., tile t_i is adjacent to tiles t_{i-1} and t_{i+1} . Initially, the robot is at tile t_1 and all tiles $t_0, \dots, t_3$ are clear (i.e., not painted). The goal of the robot is to paint tiles t_1 , t_2 , and t_3 . It can move from one tile to an adjacent one, but only if the destination tile has not been painted yet. It can paint any tile adjacent to its current position.

The following figure illustrates the initial and goal states:



Consider the following STRIPS formalization of this planning task:

- $V = \{at(x), p(x), c(x)\}$ for $x \in \{t_0, t_1, t_2, t_3\}$. Note: $at(x)$ stands for “the robot is at tile x ”, $p(x)$ stands for “tile x is painted”, and $c(x)$ stands for “tile x is clear (not painted)”.
- $A = \{move(t_x, t_y), paint(t_x, t_y)\}$ (all action costs are 1)
 - $move(t_x, t_y)$: pre: $\{at(t_x), c(t_y)\}$, add: $\{at(t_y)\}$, del: $\{at(t_x)\}$,
for $t_x, t_y \in \{t_0, t_1, t_2, t_3\}, |x - y| = 1$
 - $paint(t_x, t_y)$: pre: $\{at(t_x), c(t_y)\}$, add: $\{p(t_y)\}$, del: $\{c(t_y)\}$,
for $t_x, t_y \in \{t_0, t_1, t_2, t_3\}, |x - y| = 1$
- $I = \{at(t_1), c(t_0), c(t_1), c(t_2), c(t_3)\}$
- $G = \{p(t_1), p(t_2), p(t_3)\}$

(i) Give a shortest plan for the initial state.

(ii) Give a shortest relaxed plan for the initial state.

(iii) What is the value of $h^+(I)$?

(iv) What is the value of $h^*(I)$?

- (v) We now run one iteration of search with the h^+ heuristic. Specify the list of successor states after expanding the initial state. **For each successor, indicate the resulting state, as well as their g and h values.**

- (vi) Suppose that at some point during the search we have the following states in the open list. Which state would be selected for expansion by A^* and which state would be selected by Greedy Best First Search? If more than one state could be selected mark all of them.

State	g	h	Selected by A^*	Selected by GBFS
s_1	2	2	☐	☐
s_2	3	3	☐	☐
s_3	5	1	☐	☐
s_4	2	3	☐	☐
s_5	7	2	☐	☐
s_6	1	5	☐	☐

- (vii) Which of the following algorithms are suitable to find an optimal solution to any planning task?

	yes	no		yes	no
Greedy Best First Search with h^+	☐	☐	A^* with h^+	☐	☐
Greedy Best First Search with h^{FF}	☐	☐	A^* with h^{FF}	☐	☐

Exercise 5 (15 points)

Consider a 2-player game where Player 1 (Alice) has three possible pure strategies (A, B, or C), and Player 2 (Bob) has two possible strategies (E or F). Both players choose their strategy simultaneously, without knowing the strategy chosen by the other player. Afterwards, they earn an amount of coins corresponding to the utility shown in Table 2 (or have to pay if the utility is negative).

		Alice		
		A	B	C
Bob	E	(2, -4)	(2, -1)	(0, 0)
	F	(2, -4)	(5, -2)	(-2, 2)

Table 2: Utilities gained by each player depending on their choice of strategy. Each pair (X, Y) represents the utility gained by Alice (X) and Bob (Y).

For example, if Alice plays *B* and Bob plays *E*, then Alice wins 2 coins and Bob loses 1 coin (utility -1).

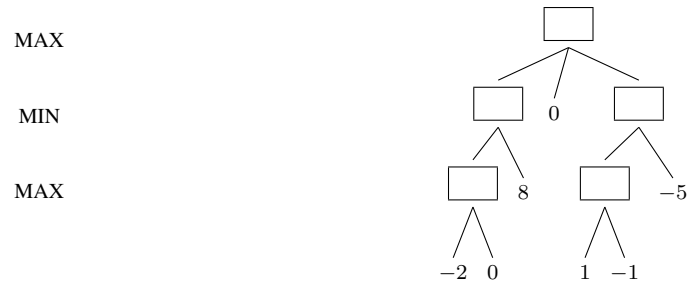
(i) Is this a zero-sum game? Justify your answer.

(ii) Consider the following pairs of strategies for Alice and Bob. Are they a Nash Equilibria? Justify your answer.

Strategy		Are they a Nash Equilibria?		
Alice	Bob	yes	no	explanation
A	E	☐	☐	
B	E	☐	☐	
C	E	☐	☐	
A	F	☐	☐	
B	F	☐	☐	
C	F	☐	☐	

(iii) Assuming that each player is trying to maximize their own utility, and both of them know that, what actions may they choose? Justify your answer.

- (iv) In the following tree, perform Minimax search, i.e., annotate all internal nodes with the correct Minimax value.



- (v) Which move should the Max player choose given the tree of part (iv)? (If there are multiple options mark all that apply)

☐ Left | ☐ Center | ☐ Right

- (vi) Assuming that both players play perfectly, who wins the game given the tree of part (iv)? (If there are multiple options mark all that apply)

☐ Max | ☐ Min | ☐ Draw | ☐ We don't know

Exercise 6 (14 points)

Consider the following probabilistic decision-making problem, modelled as a Markov Decision Process, where an agent is trying to keep in good shape. We have 4 states, $S = \{s_{injury}, s_1, \dots, s_3\}$, indicating whether the agent is injured (s_{injury}) or, otherwise, what fitness level the agent has: s_1 is the lowest level and s_3 is the highest. The starting state is s_1 and there are no terminal states. The agent has 3 actions $\{\text{relax, work-hard, work-light}\}$, whose transition probabilities are defined by the following tables.

Relax	s_{injury}	s_1	s_2	s_3	Work-hard	s_{injury}	s_1	s_2	s_3	Work-light	s_{injury}	s_1	s_2	s_3
s_{injury}	0.7				s_{injury}	1	0.2	0.2	0.2	s_{injury}	1	0.1	0.1	0.1
s'_1	0.3	1	1		s'_1					s'_1		0.5		
s'_2				1	s'_2		0.8			s'_2		0.4	0.5	
s'_3					s'_3			0.8	0.8	s'_3			0.4	0.9

Table 3: Transition probabilities $P(s, a, s')$ of each action. Each table corresponds to an action and each entry (s'_i, s_j) represents the probability of reaching s'_i after applying such an action at state s_j . Entries with 0 probability are left empty.

In words, if the agent does relax, his shape will get worse. When the agent works, it sometimes stays equally fit and sometimes improves, with a probability that depends on whether it is working hard or not. But it also has a small chance of having an injury.

The discount factor is in all cases $\gamma = 0.9$.

The agent has the following reward function:

state	s_{injury}	s_1	s_2	s_3
reward	-10	0	1	5

- (i) Suppose that the agent executes the following sequence: $s_1 \xrightarrow{\text{work-hard}} s_2 \xrightarrow{\text{work-light}} s_2 \xrightarrow{\text{relax}} s_1$. What is the value of the discounted reward obtained by the agent in this trace?

What is the probability of such a trace, given that the agent chooses those actions?

Consider the following value function V_0 (not necessarily the optimal one):

State	s_{inj}	s_1	s_2	s_3
Value V_0	-2	1	3	8

(ii) According to V_0 , what is the action that the agent should execute on state s_1 ? For each action indicate the expected value the agent expects (according to V_0), and how this is computed. Specifically, indicate intermediate operations and what numbers from the problem description are being used.

- Expected value for choosing relax on s_1 given V_0 :

- Expected value for choosing work-light on s_1 given V_0 :

- Expected value for choosing work-hard on s_1 given V_0 :

- What is the action that the agent should execute on s_1 ?

(iii) Starting with V_0 , we start to apply one iteration of value iteration. Indicate what is the new value function for s_{inj} . For each computation, clearly indicate what operation has been performed.

(iv) Starting with V_0 , we start to apply one iteration of value iteration. Indicate what is the new value function for s_1 . For each computation, clearly indicate what operation has been performed.